

TN SKILL- PROJECT REPORT
On
FARM MANGEMENT SYSTEM (FMS)

Project work submitted to



THIRUVALLUVAR UNIVERSITY – VELLORE

In partial fulfillment of the requirement

For the award of the degree of

BACHELOR OF SCIENCE

IN

DATA SCIENCE

Submitted

by

ANBUMANI M (REG NO: 20924U48002)

ARISH S (REG NO: 20924U48004)

JEEVARATHINAM M (REG NO: 20924U48017)

MOHAMED MALIK BASHA A (REG NO: 20924U48021)

UDAYAKUMAR S (REG NO: 20924U48033)



Shanmuga Industries Arts and Science College

(Affiliated to Thiruvalluvar University)

Tiruvannamalai.

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FARM MANAGEMENT SYSTEM (FMS)

S.NO	STUDENT NAME	NM ID	ROLE
1	ANBUMANI M	2459860073361A6AA774BA69F31CABE4	TEAM LEADER
2	ARISH S	034805CB42C1B32A3F14A0F646F4EB3F	MEMBER
3	JEEVARATHINAM M	0A25586644660A4AA2C4685876A54575	MEMBER
4	MOHAMED MALIK BASHA A	1D2AAB51F6D5EECB23F46A407761B7D D	MEMBER
5	UDAYAKUMAR S	2CE375D6652F60F9B61428A6FBB0AE80	MEMBER

FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

1. INTRODUCTION

1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions.

1.2 Purpose :

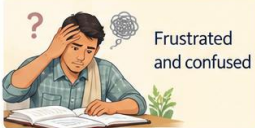
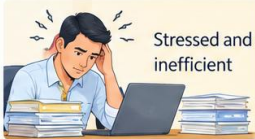
The main purpose of this project is to:

- Digitize agricultural data management
- Improve data accuracy and accessibility
- Automate record handling
- Enable real-time monitoring
- Support data-driven farming decisions
- Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	Frustrated and confused
PS-2	Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	Stressed and inefficient
PS-3	Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	Uncertain and dissatisfied
PS-4	System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	Concerned and insecure

Problem Statements – Farm Management System					
PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	 Farmer	Maintain farm and crop records	Data is often incomplete or incorrect	Records are kept manually in registers	 Frustrated and confused
PS-2	 Farm Manager	Monitor farm productivity and sales	Reports are delayed and inaccurate	Data is scattered across files and sheets	 Stressed and inefficient
PS-3	 Buyer	Access crop availability and pricing	Information is not updated regularly	There is no centralized digital system	 Uncertain and dissatisfied
PS-4	 System Admin	Manage user access and system security	Unauthorized access may occur	Access control is handled manually	 Concerned and insecure

2.2 Empathy Map Canvas

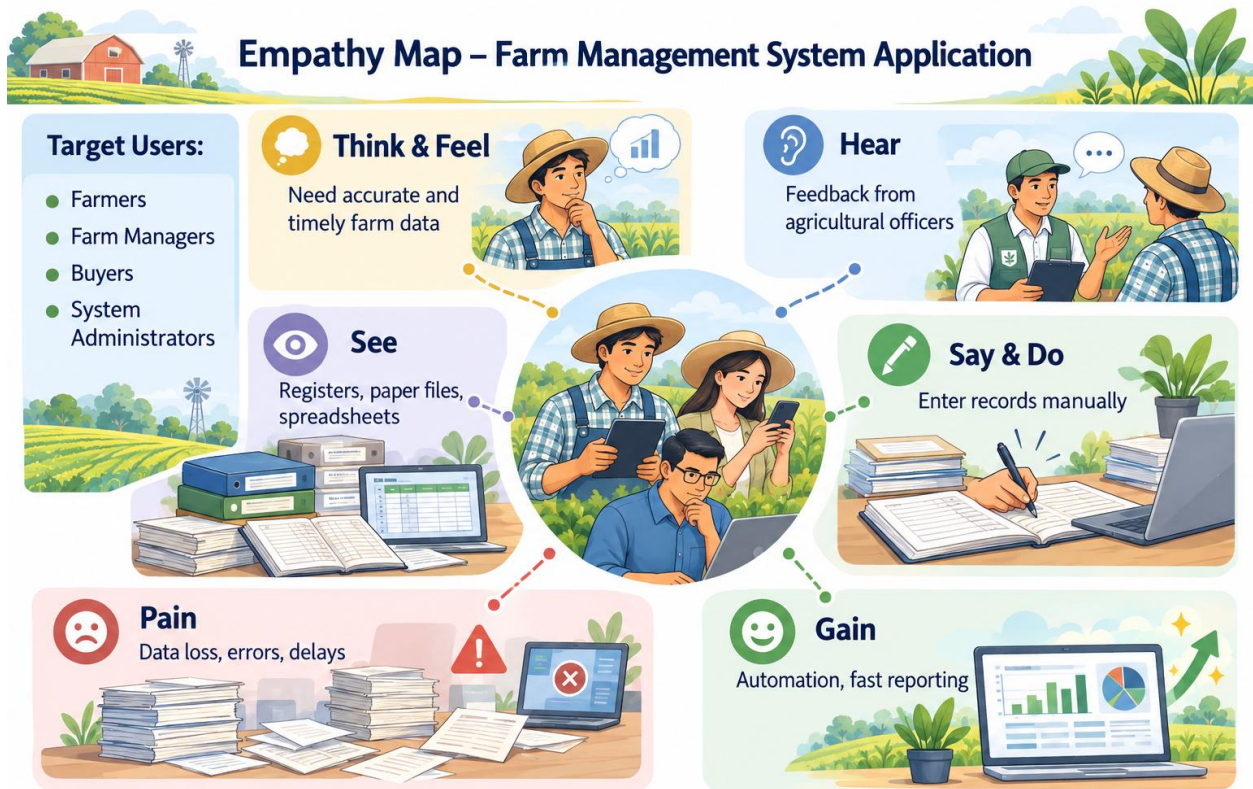
Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators

Empathy Analysis

- **Think & Feel:** Need accurate and timely farm data
- **Hear:** Feedback from agricultural officers
- **See:** Registers, paper files, spreadsheets
- **Say & Do:** Enter records manually
- **Pain:** Data loss, errors, delays

- **Gain:** Automation, fast reporting



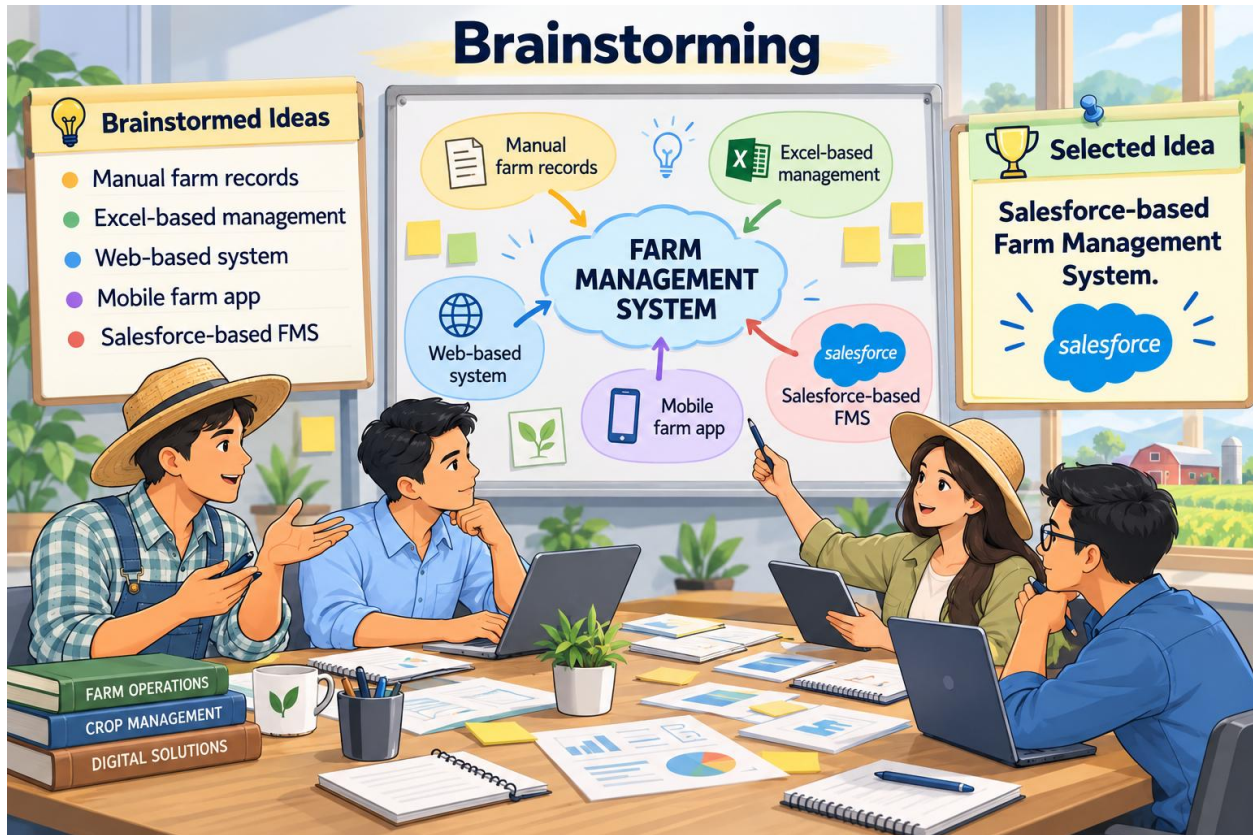
2.3 Brainstorming

Brainstormed Ideas

- Manual farm records
- Excel-based management
- Web-based system
- Mobile farm app
- Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

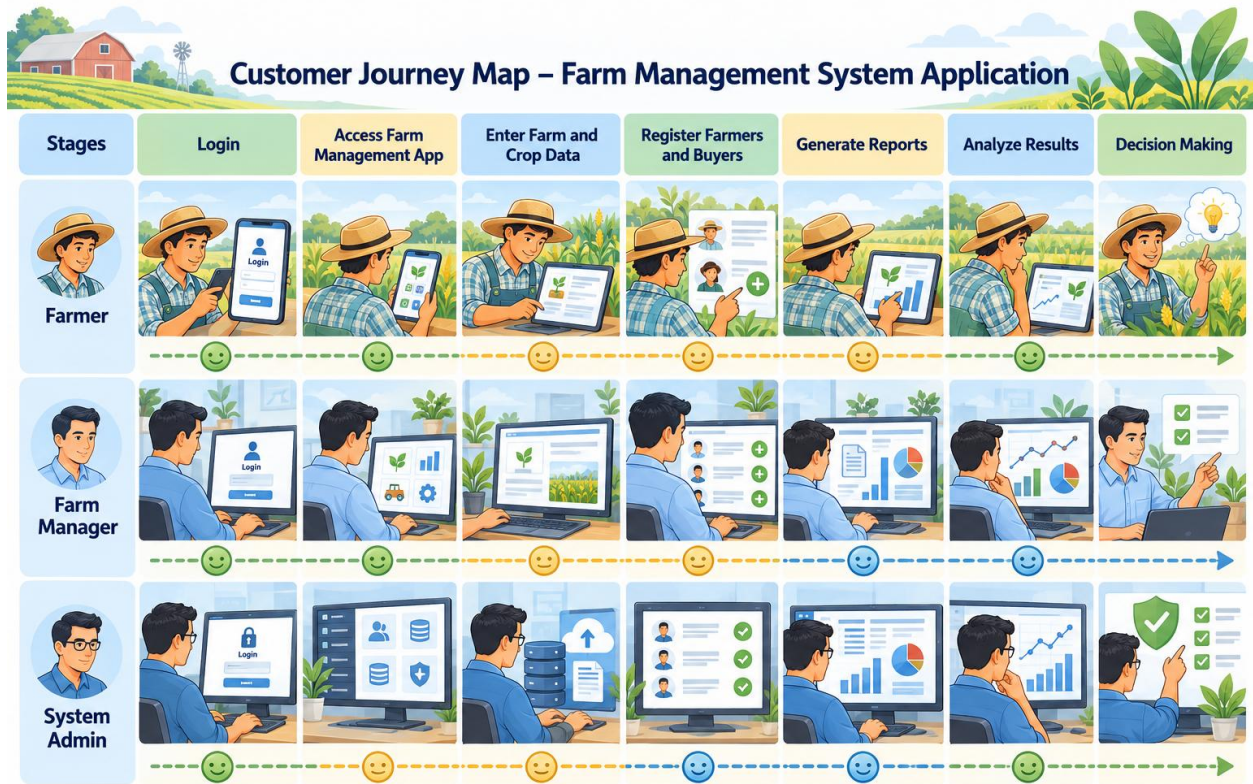


3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login
2. Access Farm Management App
3. Enter Farm and Crop Data
4. Register Farmers and Buyers
5. Generate Reports
6. Analyze Results
7. Decision Making



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation

FR-6	Dashboard Visualization
FR-7	User Role Management

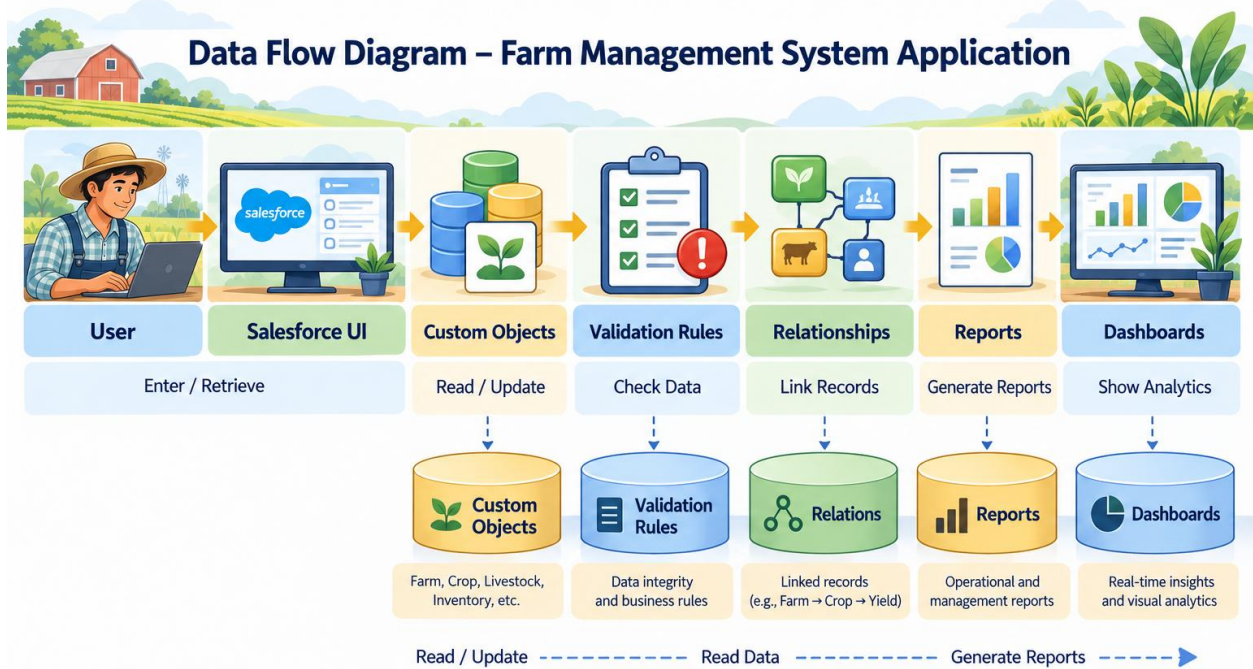
Non-Functional Requirements

NFR	Description	Example
Usability	Easy interface	Simple Lightning UI
Security	Data protection	Profiles & roles
Reliability	No data loss	Cloud backup
Performance	Fast access	Reports in <3 sec
Availability	24/7 access	Cloud hosting
Scalability	Handles growth	More farms/users

3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards



3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Validation Rules
Analytics	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management. Cloud storage prevents data loss, dashboards improve analysis, and real-time reports eliminate delays. This ensures accurate and timely agricultural insights.



4.2 Proposed Solution

Functional Requirements

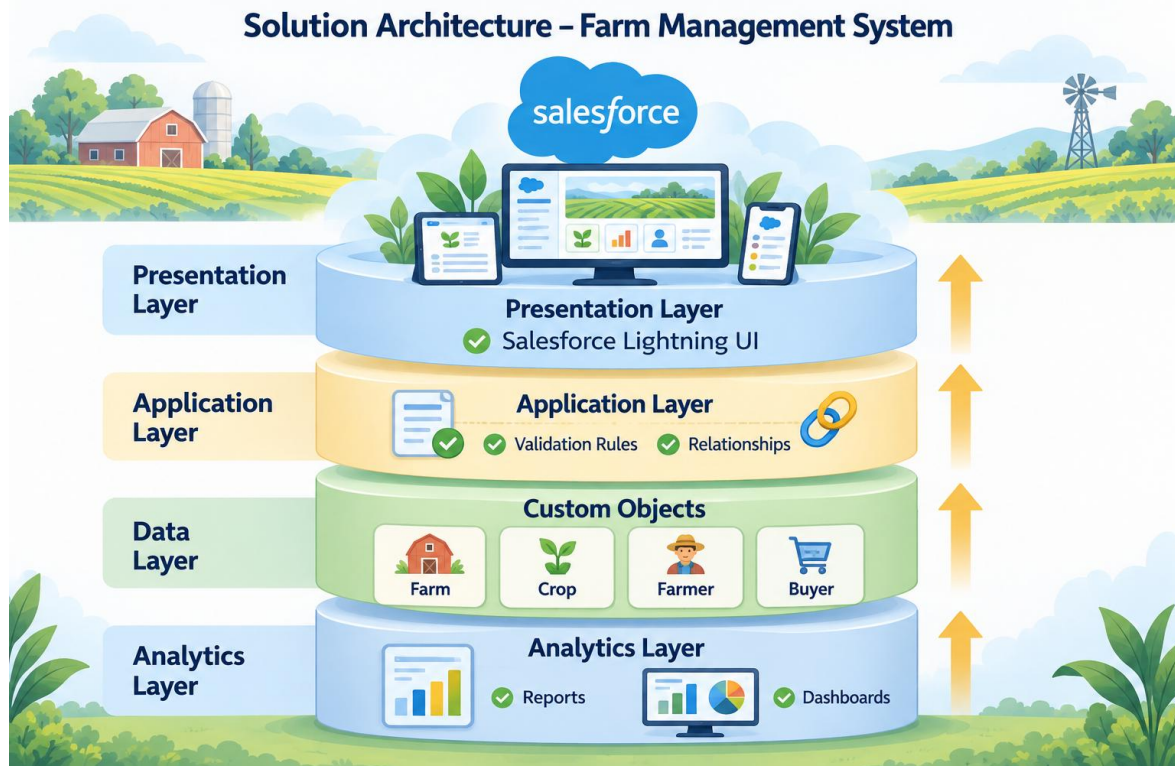
Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.

Solution	Salesforce-based Farm Management System with centralized data storage and automation.
Innovation	Automated data entry validation, cloud storage, and real-time reporting dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and managers.es
Business Model	Digital farm management and agricultural data services platform.
Scalability	Supports multiple farms, users, and regions with cloud-based infrastructure.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI
2. Application Layer – Validation Rules, Relationships
3. Data Layer – Custom Objects
4. Analytics Layer – Reports and Dashboards



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.	3	High	Member1
Sprint-2a	Data Modeling	USN-2	As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.	3	High	Member4
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.	5	High	Member2
Sprint-5	Reports	USN-7	As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

6. Project Development Phase:

Project Description:

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

Milestone 1-Salesforce:

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

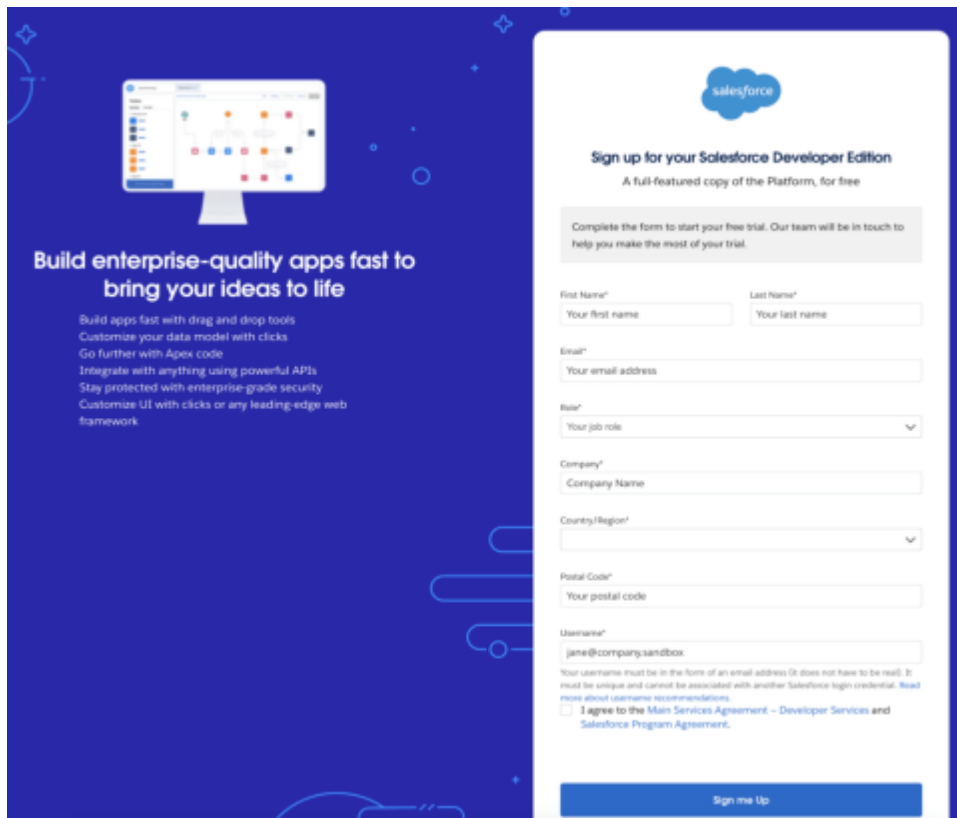
Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity1:

Creating a Salesforce Developer Org:

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search [Developer.salesforce.com/signup](https://developer.salesforce.com/signup)
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



Build enterprise-quality apps fast to bring your ideas to life

Build apps fast with drag and drop tools
Customize your data model with clicks
Go further with Apex code
Integrate with anything using powerful APIs
Stay protected with enterprise-grade security
Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name*
Your first name

Last Name*
Your last name

Email*
Your email address

Role*
Your job role

Company*
Company Name

Country/Region*
Country/Region

Postal Code*
Your postal code

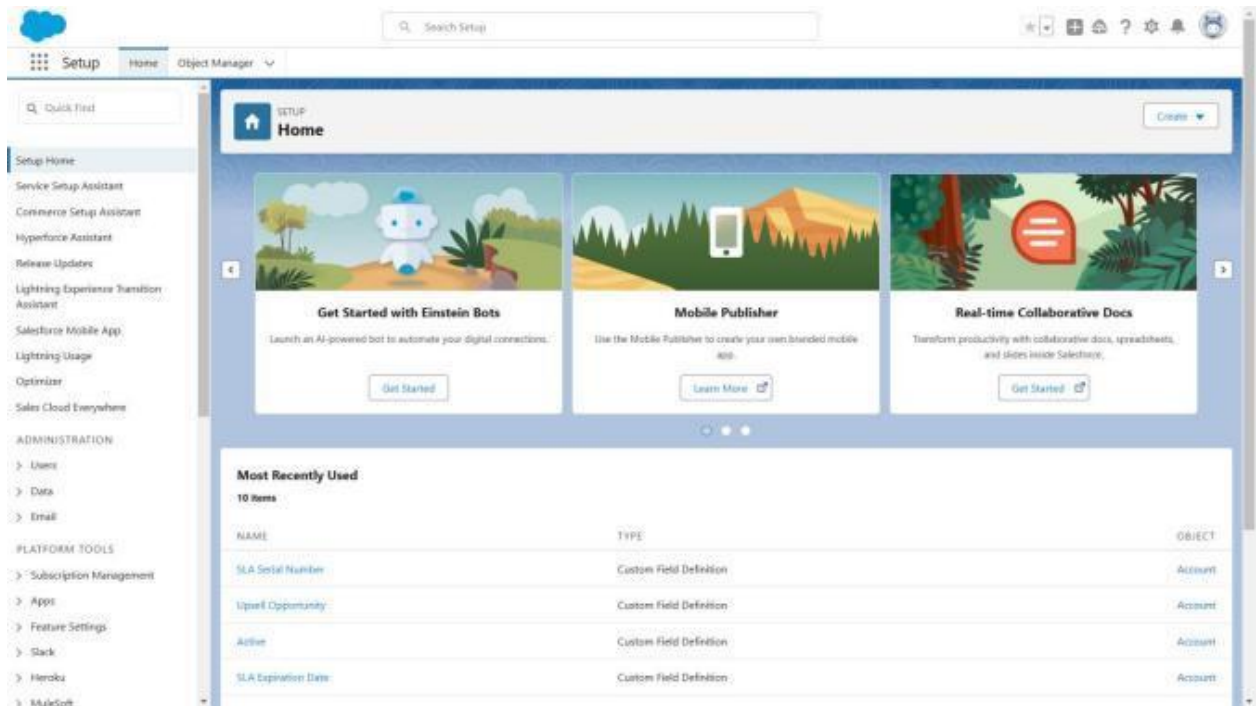
Username*
jane@company.sandbox

Your username must be in the form of an email address (it does not have to be real). It must be unique and cannot be associated with another Salesforce login credential. [Read more about username recommendations.](#)

☐ I agree to the [Main Services Agreement – Developer Services](#) and [Salesforce Program Agreement](#).

Sign me Up

6. By using username and password you can into the Salesforce org. The homepage will appear below



Milestone 2-Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

In This Application We Use 4 Custom Objects:

- 1.Farm
- 2.Crop
- 3.Farmer
- 4.Buyer

Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.
2. Click create, select custom object.



3. Fill in the label as " Farm ".
4. Fill in the plural label as " Farms".
5. Record name: Farm Name"
6. Select the data type as "Text".

The screenshot shows the Salesforce Object Setup page for an object named 'Farm'. The left sidebar contains a navigation menu with the following items: Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled 'Data Type' and shows 'Text' selected. A warning message states: 'Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.' Below this, there are four sections: 'Optional Features' with checkboxes for 'Allow Reports' (checked), 'Allow Actions' (unchecked), 'Track Field History' (unchecked), 'Allow in Chatter Groups' (unchecked), and 'Enable Licensing' (unchecked); 'Object Classification' with a note about Enterprise vs. Lightning Application objects and checkboxes for 'Allow Sharing' (checked), 'Allow Bulk API Access' (checked), and 'Allow Streaming API Access' (checked); 'Deployment Status' with radio buttons for 'In Development' (unchecked) and 'Deployed' (checked); and 'Search Status' with a note about search results and a checkbox for 'Allow Search' (checked). At the bottom right, there is a 'What is this?' link. At the bottom center, there are three buttons: 'Save', 'Save & Show', and 'Cancel'.

Data Type: **Text** Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Optional Features

- ☒ Allow Reports
- ☐ Allow Actions
- ☐ Track Field History
- ☐ Allow in Chatter Groups
- ☐ Enable Licensing [?](#)

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Lightning Application object. [Learn more](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status [What is this?](#)

- ☐ In Development
- ☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more](#)

- ☒ Allow Search

[Save](#) [Save & Show](#) [Cancel](#)

7. In the Optional Features section, select Allow Reports and Track Field History.
 8. In the Deployment Status section, ensure Deployed is selected.
 9. In the Search Status section, select Allow Search.
 10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
-
11. Leave everything else as is, and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox
 - g. Check the Allow Search checkbox
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout
 - i. Click Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Farmer

- c. Plural Label: Farmers
- d. Record Name: Farmer Name e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Buyer
 - c. Plural Label: Buyers
 - d. Record Name: Buyer Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
 - h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
 - i. Click Save.

Milestone 3 -

Tab: Tab

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

Salesforce Setup interface showing the Custom Tabs configuration page.

Left Sidebar:

- Setup
- Home
- Object Manager
- Search Setup
- tabs
- User Interface
- Rename Tabs and Labels
- Tab** (highlighted with a red box)

Main Content Area:

Custom Tabs

You can create new custom tabs to extend Salesforce functionality or to build new application functionality.

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Custom Object Tabs (New button highlighted with a red box)

Action	Label	Tab Style	Description
Edit Del	Devs	Form	
Edit Del	Candidates	Form	
Edit Del	Cases	Form	
Edit Del	Enrollments	Form	
Edit Del	Enrollments	Form	
Edit Del	Job Applications	Form	
Edit Del	Jobs	Form	
Edit Del	Leads	Form	
Edit Del	Recruiters	Form	
Edit Del	Students	Form	

Web Tabs (New button highlighted with a red box)

2. Under custom object tabs, click New.

The screenshot shows the Salesforce Setup interface for creating a new custom object tab. The left sidebar contains the Setup menu with options like Home, Object Manager, and Tabs. The main content area is titled "New Custom Object Tab" and includes a "Step 1. Enter the Details" section. A red box highlights the "Tab Style" dropdown menu, which is currently set to "None". Below this, there is a "Description" text area and a "Next" button.

Setup

Home Object Manager

tabs

User Interface

Rename Tabs and Labels

Tabs

Didn't find what you're looking for? Try using Global Search.

SETUP Tabs

New Custom Object Tab

Step 1. Enter the Details

Step 1 of 3

Choose the custom object for this new custom tab. Fill in other details.

New Custom Object Tab

Required Information

Select an existing custom object or create a new custom object now.

Object: None

Tab Style: None

(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link: None

Enter a short description.

Description

Next Cancel

3. For Object, select Farm.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.

4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

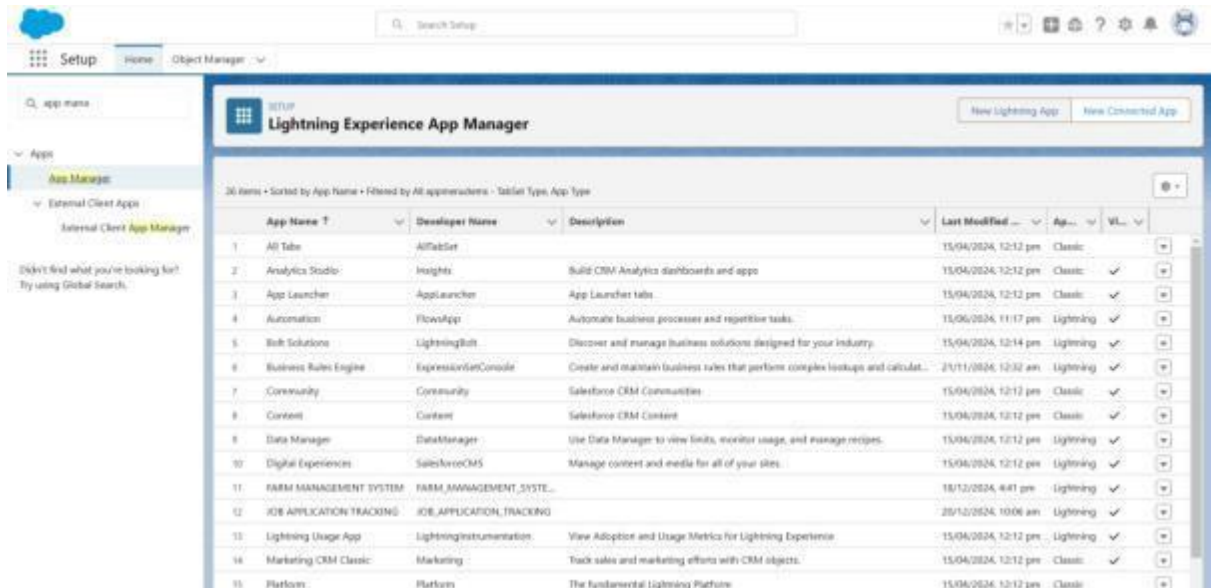
Milestone 4 - Lightning App:

What is an App?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

Activity 1:

Create the Farm Management System app



The screenshot shows the Salesforce Lightning Experience App Manager interface. The left sidebar contains navigation links for Setup, Home, and Object Manager. The main content area displays a list of 26 installed apps, sorted by App Name. The table includes columns for App Name, Developer Name, Description, Last Modified, App Type, and a checkbox for visibility. The 'Farm Management System' app is listed as item 11.

App Name	Developer Name	Description	Last Modified	App Type	Visible
1 All Tabs	AllTabs		11/04/2024, 12:12 pm	Classic	
2 Analytics Studio	Insights	Build CRM Analytics dashboards and apps	11/04/2024, 12:12 pm	Classic	✓
3 App Launcher	AppLauncher	App Launcher tabs	11/04/2024, 12:12 pm	Classic	✓
4 Automation	FlowApp	Automate business processes and repetitive tasks	11/06/2024, 11:17 pm	Lightning	✓
5 Bulk Solutions	LightningBolt	Discover and manage business solutions designed for your industry	11/04/2024, 12:14 pm	Lightning	✓
6 Business Rules Engine	ExpressionEngine	Create and maintain business rules that perform complex lookups and calculations	21/11/2024, 12:02 am	Lightning	✓
7 Community	Community	Salesforce CRM Communities	11/04/2024, 12:12 pm	Classic	✓
8 Content	Content	Salesforce CRM Content	11/04/2024, 12:12 pm	Classic	✓
9 Data Manager	DataManager	Use Data Manager to view limits, monitor usage, and manage recipes	11/04/2024, 12:12 pm	Lightning	✓
10 Digital Experiences	SalesforceCMS	Manage content and media for all of your sites	11/04/2024, 12:12 pm	Lightning	✓
11 FARM MANAGEMENT SYSTEM	FARM_MANAGEMENT_SYSTE...		18/12/2024, 4:41 pm	Lightning	✓
12 JOB APPLICATION TRACKING	JOB_APPLICATION_TRACKING		20/12/2024, 10:06 am	Lightning	✓
13 Lightning Usage App	LightningInstrumentation	View Adoption and Usage Metrics for Lightning Experience	11/04/2024, 12:12 pm	Lightning	✓
14 Marketing CRM Classic	Marketing	Track sales and marketing efforts with CRM objects	11/04/2024, 12:12 pm	Classic	✓
15 Platform	Platform	The fundamental Lightning Platform	11/04/2024, 12:12 pm	Classic	

- From Setup, enter App Manager in the Quick Find and select App Manager.
 - Click New Lightning App. Farm Management System as the App Name, then click Next
 - Under App Options, leave the default selections and click Next.
 - Under Utility Items, leave as is and click Next.
 - From Available Items, select Farm, Crop,Farmer,Buyer Reports,and Dashboards and move them to Selected Items. Click Next. ● From Available Profiles, select System Administrator and move it to Selected Profiles.Click Save & Finish.
 - To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.
- Note:
1. App Launcher-Displays available apps.
 2. App Name-Displays the current selected app.

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name Name your app...

* Developer Name Enter a developer name...

Description Enter a description...

App Branding

Image Upload

Primary Color Hex Value: #0070C2

Org Theme Options ☐ Use the app's image and color instead of the org's custom theme

App Launcher Preview

Next

13	Lightning Usage App	Lightning instrumentation	View Adoption and Usage Metrics for Lightning Experience	15/04/2024, 12:12 pm	Lightning	✓
14	Marketing CRM Classic	Marketing	Track sales and marketing efforts with CRM objects.	15/04/2024, 12:12 pm	Classic	✓
15	Platform	Platform	The fundamental Lightning Platform	15/04/2024, 12:12 pm	Classic	

3. Navigation menu-Displays the tabs available inside the app.

The screenshot displays the Lightning App Builder interface for a 'FARM MANAGEMENT SYSTEM'. The left sidebar shows the 'App Settings' menu with 'Navigation Items' selected. The main content area is titled 'Navigation Items' and includes a descriptive paragraph: 'Choose the items to include in the app, and arrange the order in which they appear. Users can personalize the navigation to add or move items, but users can't remove or rename the items that you add. Some navigation items are available only for phone or only for desktop. These items are dropped from the navigation bar when the app is viewed in a format that the item doesn't support.'

The interface is divided into two main sections: 'Available Items' and 'Selected Items'.

Available Items: A list of items that can be added to the navigation menu, including Accounts, All Sites, Alternative Payment Methods, Analytics, App Launcher, Appointment Categories, Appointment Invitations, Approval Requests, Asset Action Sources, Asset Actions, Asset State Periods, and Assets. A search bar at the top of this list is labeled 'Type to filter list...'. A 'Create' button is located at the top right of the list.

Selected Items: A list of items that have been added to the navigation menu. The items listed are Farms, Crops, Farmers, Buyers, Reports, and Dashboards. This section is highlighted with a red border in the image. Arrows on the right side of this list allow for reordering the items.

Milestone 5 -

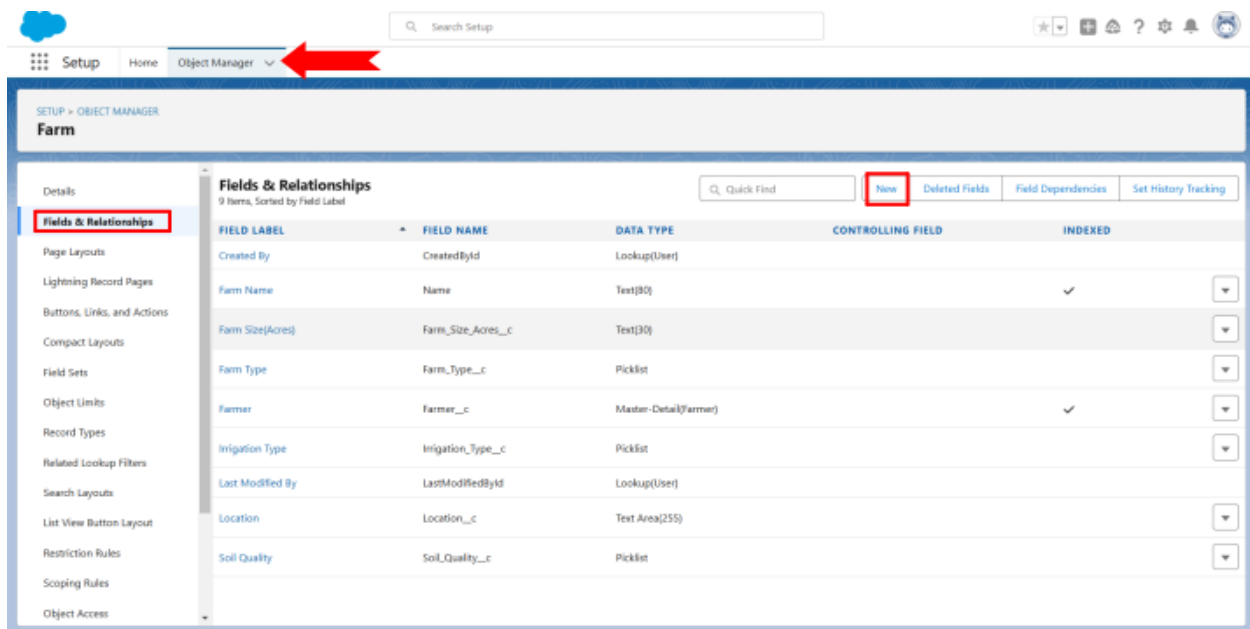
Fields and Relationship: What are fields?

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New



The screenshot shows the Salesforce Setup interface. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. A red arrow points to the 'Object Manager' tab. The left sidebar shows the 'Fields & Relationships' section highlighted. The main content area displays the 'Fields & Relationships' section for the 'Farm' object, with a table listing fields and their properties.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Farm Name	Name	Text(80)		✓
Farm Size(Acres)	Farm_Size_Acres__c	Text(30)		
Farm Type	Farm_Type__c	Picklist		
Farmer	Farmer__c	Master-Detail(Farmer)		✓
Irrigation Type	Irrigation_Type__c	Picklist		
Last Modified By	LastModifiedById	Lookup(User)		
Location	Location__c	Text Area(255)		
Soil Quality	Soil_Quality__c	Picklist		

5. Choose Data type as a picklist, Click next

The screenshot shows the Salesforce Setup interface for the 'Farm' object. The 'Fields & Relationships' section is active, displaying a list of data types. The 'Picklist' option is selected and highlighted with a red box. The 'Next' button at the bottom right is also highlighted with a red box.

Fields & Relationships

- ☐ Checkbox
- ☐ Currency
- ☐ Date
- ☐ Date/Time
- ☐ Email
- ☐ Geolocation
- ☐ Number
- ☐ Percent
- ☐ Phone
- ☒ **Picklist**
- ☐ Picklist (Multi-Select)
- ☐ Text
- ☐ Text Area
- ☐ Text Area (Long)
- ☐ Text Area (Rich)
- ☐ Text (Encrypted)
- ☐ Time
- ☐ URL

Next Cancel

6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)
7. Next, Next Click Save.

The screenshot shows the Salesforce Setup interface with the 'Object Manager' tab selected. The 'Farm' object is chosen, and the 'New Custom Field' wizard is open. The 'Field Label' is 'Farm Type' and the 'Field Name' is 'Farm_Type'. The 'Values' are 'Crop', 'Livestock', and 'Mixed'. The 'Field Name' is 'Farm_Type'. The 'Field Label' is 'Farm Type'. The 'Values' are 'Crop', 'Livestock', and 'Mixed'. The 'Field Name' is 'Farm_Type'. The 'Field Label' is 'Farm Type'. The 'Values' are 'Crop', 'Livestock', and 'Mixed'. The 'Field Name' is 'Farm_Type'.

Field Label: Farm Type

Values: ☐ Use global picklist value set ☒ Enter values, with each value separated by a new line

Crop
Livestock
Mixed

☐ Display values alphabetically, not in the order entered
☐ Use first value as default value
☒ Restrict picklist to the values defined in the value set

Field Name: Farm_Type

Description:

Help Text:

Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

- 1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values (Organic,Cover Crops,Composting)
- 2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler irrigation,Drip irrigation,localized irrigation)
- 3.Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter field label (size)
- 5.Create a Master Detail relationship with Farmer Object.

Activity 2:

Creation of fields for the Crop objects:

- 1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save. 2.Select Data type as date ,Enter Field label (Planting Date),then save .
- 3.Again we select data type as a date,Enter field labe (Harvest date),save
- 4.Next We Create Lookup relationship on crop object realted to Farm Object.
- 5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it.
- 6.Select Data type as a Number,enter Field Label (Total Acres Harvested),Enter Maximum Length(20) and save.
- 7.Now We create Formula Field For That Select data Type As a Formula,Click Next.



Search Setup



Setup Home Object Manager

Setup > OBJECT MANAGER
Crop

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected

Select one of the data types below.

☐ Auto Number

A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

☒ Formula

A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

☐ Roll-Up Summary

A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

☐ Lookup Relationship

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

☐ Master-Detail Relationship

Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where:

- The relationship field is required on all detail records.
- The ownership and sharing of a detail record are determined by the master record.
- When a user deletes the master record, all detail records are deleted.
- You can create rollup summary fields on the master record to summarize the detail records.

The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.

☐ External Lookup Relationship

Creates a relationship that links this object to an external object whose data is stored outside the Salesforce org.

☐ Checkbox

Allows users to select a True (checked) or False (unchecked) value.

☐ Currency

Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.

☐ Date

Allows users to enter a date or pick a date from a popup calendar.

☐ Date/Time

Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the Date/Time field.

☐ Email

Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.

8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

The screenshot shows the Salesforce Setup interface for the 'Crop' object. The left sidebar contains navigation links: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled 'New Custom Field' and is at 'Step 3 of 6: Enter formula'. It includes a 'Previous' button, a 'Next' button, and a 'Cancel' button. Below the navigation buttons, there is a text input field for the formula, with an example: 'Gross Margin = Amount - Cost'. To the right of the formula input is an 'Insert Field' button. Below the formula input, there is a red rectangular box highlighting the formula 'Actual_Yield__c / Total_Acre_Harvested__c'. To the right of the formula input is an 'Insert Operator' button. On the far right, there is a 'Functions' panel with a dropdown menu for 'All Function Categories' and a list of functions: ABS, ACOS, ADDMONTHS, AND, ASCII, and ASIN. Below the list is an 'Insert Selected Function' button.

Activity 3:

Creation of fields for the Farmer objects:

1. Select Data type Phone, Enter field label (Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamilnadu), save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer object, select Fields & Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana - Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the associate Values available for Particular state.



Search Setup



Setup

Home

Object Manager

Farmer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

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New Field Dependency

[Help for this Page](#)

Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.

- The field that drives filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 200 values can be controlling fields.
- The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.

Step 1. Select a controlling field and a dependent field. Click Continue when finished.

Step 2. On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.

Controlling Field	--None--	Continue	Cancel
Dependent Field	--None--	Continue	Cancel



SETUP

Dependent Field	District
-----------------	----------

▼ Instructions

- Double click on a cell to toggle its visibility for the Controlling Field value shown in the column heading.
- To change multiple cells at once, select multiple cells and then click the Include Values or Exclude Values button to change the visibility of all selected cells at once.
- Use SHIFT + click to select a range of adjacent cells. Use CTRL + click to select multiple cells that are not adjacent.
- Use the Preview button to test the results.

Legend

Excluded Value

Included Value

Click button to include or exclude selected values from the dependent picklist:

Include Values

Exclusive Values

Showing Columns: 1 - 3 (of 3) < Previous | Next > [View All](#) [Go to](#)

State: District:	Telangana	Andhra Pradesh	Tamil Nadu
	Taddei Bhongir	Vadali Bhongir	Vadali Bhongir
	Nalgonda	Nalgonda	Nalgonda
	Warangal	Warangal	Warangal
	Vizag	Vizag	Vizag
	Kurnool	Kurnool	Kurnool
	Chittoor	Chittoor	Chittoor
	Chennai	Chennai	Chennai
	Coimbatore	Coimbatore	Coimbatore
	Dharmapuri	Dharmapuri	Dharmapuri

Showing Columns: 1 - 3 (of 3) < Previous | Next > View All

[Click here to see full-size, printable, color-coded version. Save the changes and print!](#)

☐ Include Values

Exclude Values

Save

☐ Cancel[Preview](#)

Save Cancel Preview

7. Click Close to close the preview window, Click Save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label(Contact), Save.
2. Select Data type as an Email, Enter Field Label(Email), Next save.
3. Select Data type Text area, Enter Field Label(Address), Save it.
4. Select Data type Picklist, Field Label(Buyer Type), Enter values (Wholesale, Retail), Click Save.

Milestone 6- Profile:

What is a profile?

A profile is a group/collection of settings and permissions that define what a user can do in

salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup , enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.



Search Setup



Setup

Home

Object Manager

Q profile

Users

Profiles

Didn't find what you're looking for?
Try using Global Search.

Setup

Profiles

Help for this Page

All Profiles

1

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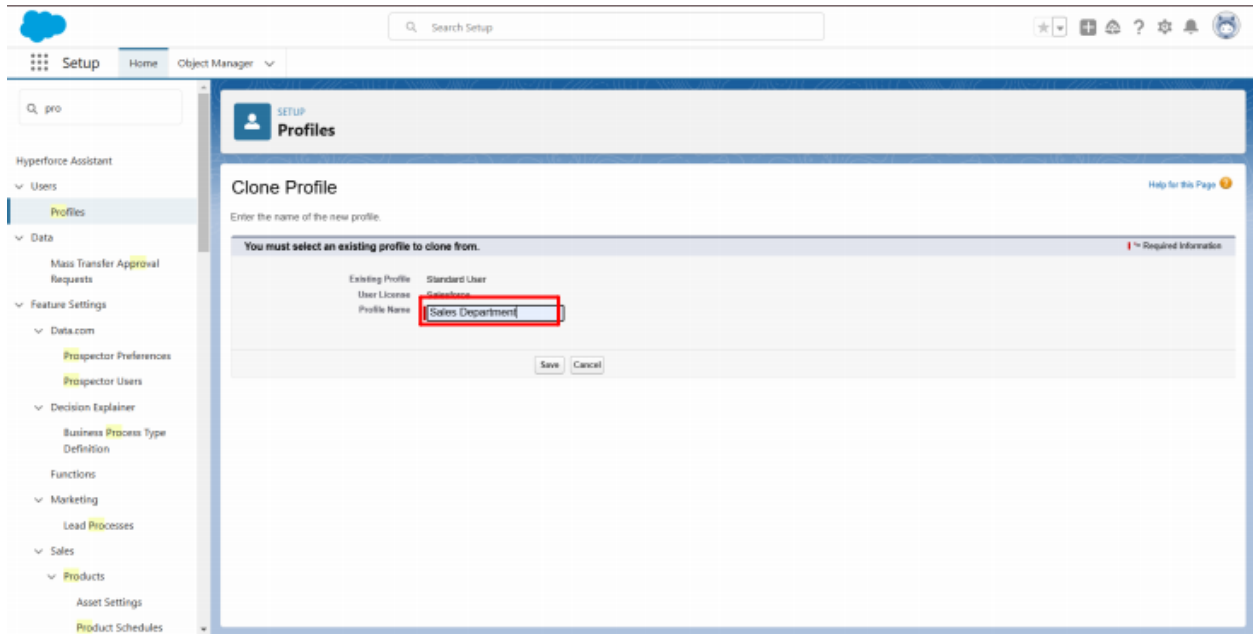
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Action	Profile Name	User License	Custom
☐ Edit Del ...	Sales Development	Salesforce Platform	☒
☐ Edit Del ...	Sales Manager	Salesforce	☒
☐ Edit Del ...	Salesforce API Core System Integrations	Salesforce Integration	☒
☐ Edit Clone	Silver Partner User	Silver Partner	☐
☐ Edit Clone	Subsidiary Manager	Salesforce	☐
☐ Edit Clone	Standard Platform User	Salesforce Platform	☐
☐ Edit Del ...	Standard Platform user	Salesforce Platform	☒
☐ Edit Clone	Standard User	Salesforce	☐
☐ Edit Clone	System Administrator	Salesforce	☐



5. Click save.

Activity 2:

Create a profile with the profile name as “ Farm Manager” .

6. From setup , enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

Milestone 7- Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

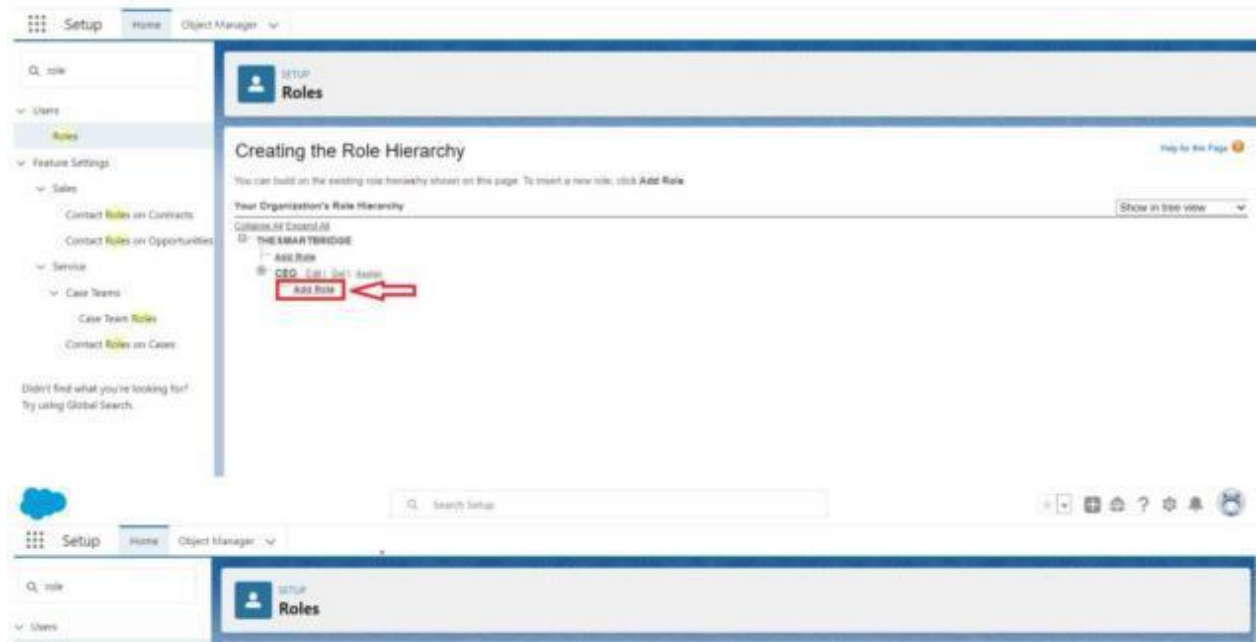
Activity-1

Creation of role

1. From the Quick find box search for the role and click on the roles option
2. select the set up roles option



3. Below the CEO click on add role and enter the label name as a " Farm Manager " and role name will be Automatically populated and click on save.



Milestone 8-User :

What is a user?

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you don't find salesforce license then deactivate a user who has salesforce license Or change the license type from Salesforce to any other.

Activity 1:

To Create a user:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

The screenshot shows the Salesforce Setup interface. On the left, the 'Setup' menu is open, and 'Users' is selected under 'User Management Settings'. The main content area is titled 'All Users' and contains a table of existing users. The 'New User' button is highlighted with a red box. Below is a table of users:

Action	Full Name	Alias	Username	Role	Action	Profile
☐ Edit	Developer1	dev1	sashanth2005@gmail.com	CEO	☐	Standard Platform user
☐ Edit	Developer2	dev2	sashanthnagaraj73@gmail.com	COO	☐	Standard Platform user
☐ Edit	Manager_HR	hrman	hrman@example.com	HR Manager	☐	Standard Platform User
☐ Edit	Jeffrey Sutherland	JSuth	chrissutherland@salesforce.com	CEO	☒	System Administrator
☐ Edit	User Integration	intgr	integration@00ae50002ce4c2ap.com		☒	Analytics Cloud Integration User
☐ Edit	Chatter Expert	chatter	chatter.00ae50002ce4c2ap.com@chatter.salesforce.com		☒	Chatter Free User
☐ Edit	User Security	sec	security@00ae50002ce4c2ap.com		☒	Analytics Cloud Security User
☐ Edit Login	Manager Sales	smann	smann@example.com		☒	Sales Manager
☐ Edit Login	Manager Farm	farmm	farmm@gmail.com	Farm Manager	☒	Farm Manager
☐ Edit Login	Person Sales	person	person@example.com		☒	Sales Consultant

3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile(Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as "Sales Department", and assign him the sales Manager profile. Follow the steps from above Activity

Milestone 9-Sharing Rules:

What are Sharing Rules?

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

- Owner-based Sharing Rules
- Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

- 1) Goto Sharing Settings, which can be found under the Quick Find section.
- 2) Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
- 3) Add the label of the sharing rule you want to make.
- 4) Select your rule type based on the criteria.
- 5) Select the field can join immediately check field from the Crop object.
- 6) Select the Crop as equal and value is True.
- 7) And in selecting the users to share with the section select roles and in that select Farm Manager.
- 8) And in the section of select the level of access for the users give the access Read/Write.
- 9) And save the rule.

Setup

Home

Object Manager

Search Setup

Q sharing

Security

Guest User

Sharing Rule Access Report

Sharing Settings

Didn't find what you're looking for? Try using Global Search.

Sharing Settings

Use sharing rules to make automatic exceptions to your organization-wide sharing settings for defined sets of users.
Note: "Roles and subordinates" includes all users in a role, and the roles below that role.
You can use sharing rules only to grant wider access to data, not to restrict access.

Label Crop

Rule Name Crop

Description

Step 1: Select your rule type

Criteria

Field	Operator	Value	
Crop Name	equals	true	AND
-None-	-None-		AND
-None-	-None-		AND
-None-	-None-		AND
-None-	-None-		AND

Add Other Criteria

Include records owned by users who can't have an assigned role

Share with Rule: Farm Manager

Access Level ReadWrite

Created by Subash Jaiswal 18/12/2024, 2:29 pm

Modified by Subash Jaiswal 19/12/2024, 2:29 pm

Save Cancel

Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record(Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

The screenshot shows the 'Edit Kesava' form in the Farm Management application. The form is titled 'Edit Kesava' and has a 'Required Information' icon. The fields are as follows:

- * Farmer Name:** Kesava
- Owner:** Sudarsh Jilagam
- Contact Number:** 796262776
- State:** Andhra Pradesh
- District:** Chittoor
- Village:** Vindela
- Farm Size(Acres):** 15

At the bottom of the form, there are three buttons: 'Cancel', 'Save & New', and 'Save'. The 'Save' button is highlighted with a red box.

Milestone 11-Reports:

What are Reports?

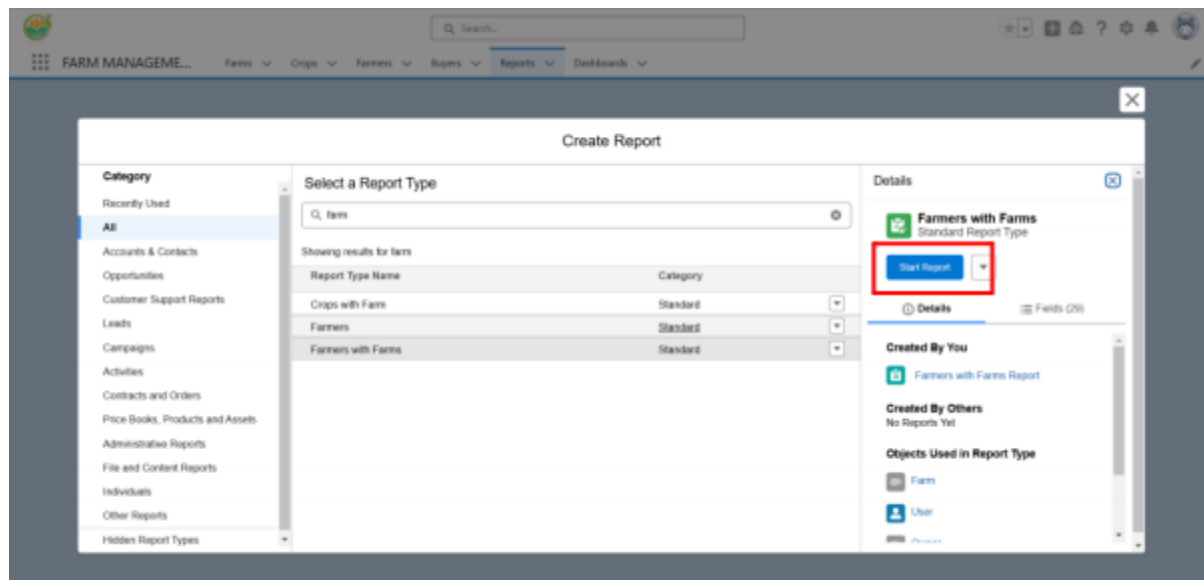
A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is stored in a folder. Folders can be public, hidden, or shared, and can be set to read-only or read/write.

Activity 1:

Create a report:

Create a report that displays rating of the account and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group,Select Which Fields You Want to Group.
- 6) Save the report by giving label name(Farmer with Farms) and save the folder as a public folder and save the report.

The screenshot shows the 'FARM MANAGEMENT' interface with the 'Reports' tab selected. The report is titled 'Farmers with Farms Report' and is currently in 'Preview' mode. The report displays a table with the following columns: Farmer: Farmer Name, Farm: Farm Name, Contact Number, Farm Size(Acres), and Farm Type. The data is grouped by 'Irrigation Type' (Localized Irrigation (3)). The table shows three rows of data: Paddy Farm, Sugar Farm, and Wheat Farm. The total number of records is 25.

Irrigation Type	Farmer: Farmer Name	Farm: Farm Name	Contact Number	Farm Size(Acres)	Farm Type
Localized Irrigation (3)	Rajesh	Paddy Farm	988438778	10	Livestock
	Rajesh	Sugar Farm	988438778	10	Crop
	Kishu	Wheat Farm	788202778	15	Livestock
Subtotal				25	
Total (3)				25	

The interface also includes a sidebar with 'Groups' and 'Columns' sections. The 'Columns' section shows the following fields selected: Farmer: Farmer Name, Farm: Farm Name, Contact Number, Farm Size(Acres), and Farm Type. The 'Groups' section shows 'Irrigation Type' selected. At the bottom, there are checkboxes for 'Row Counts', 'Detail Rows', 'Subtotals', and 'Grand Total', all of which are checked.

Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

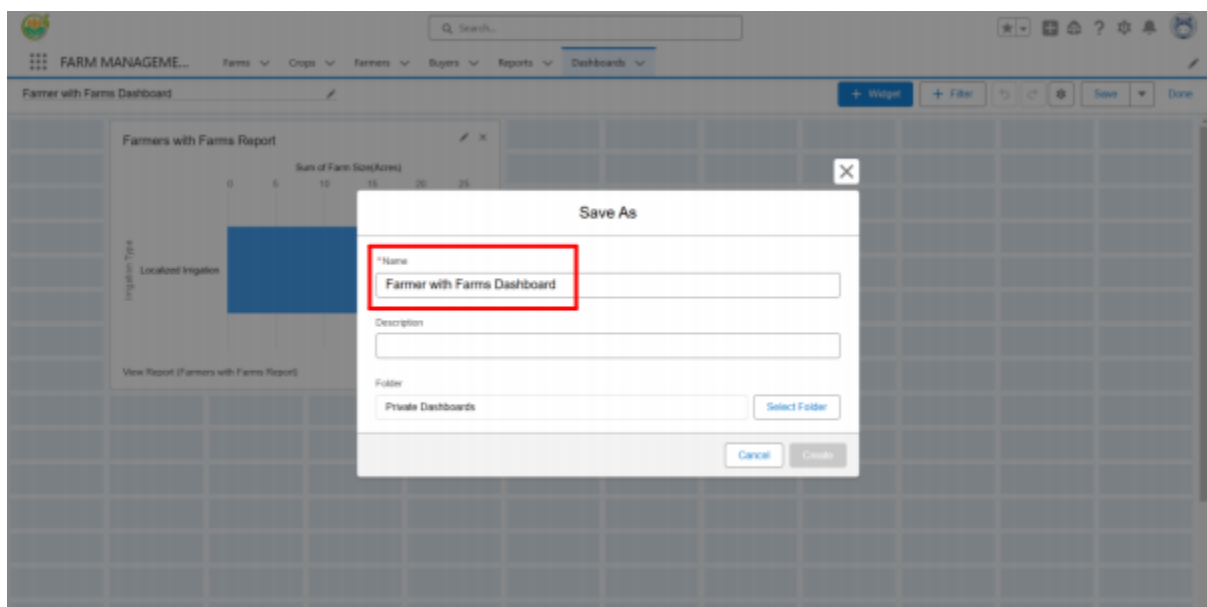
Milestone 12-Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

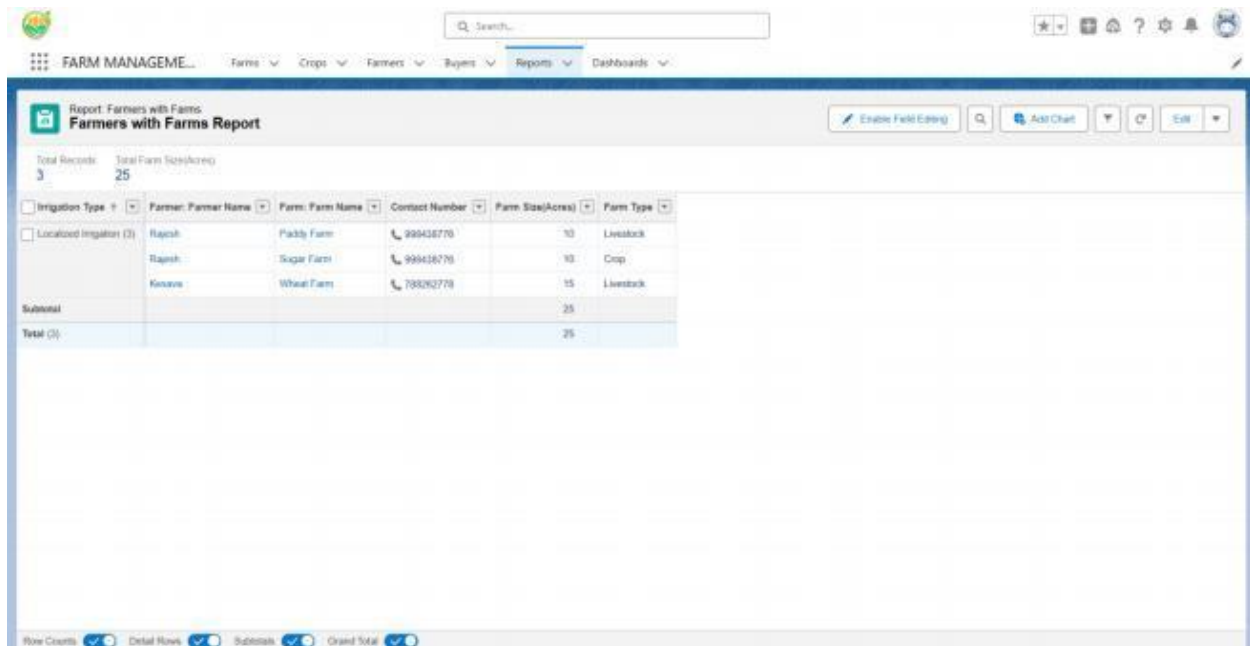
1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



Milestone-13: View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records

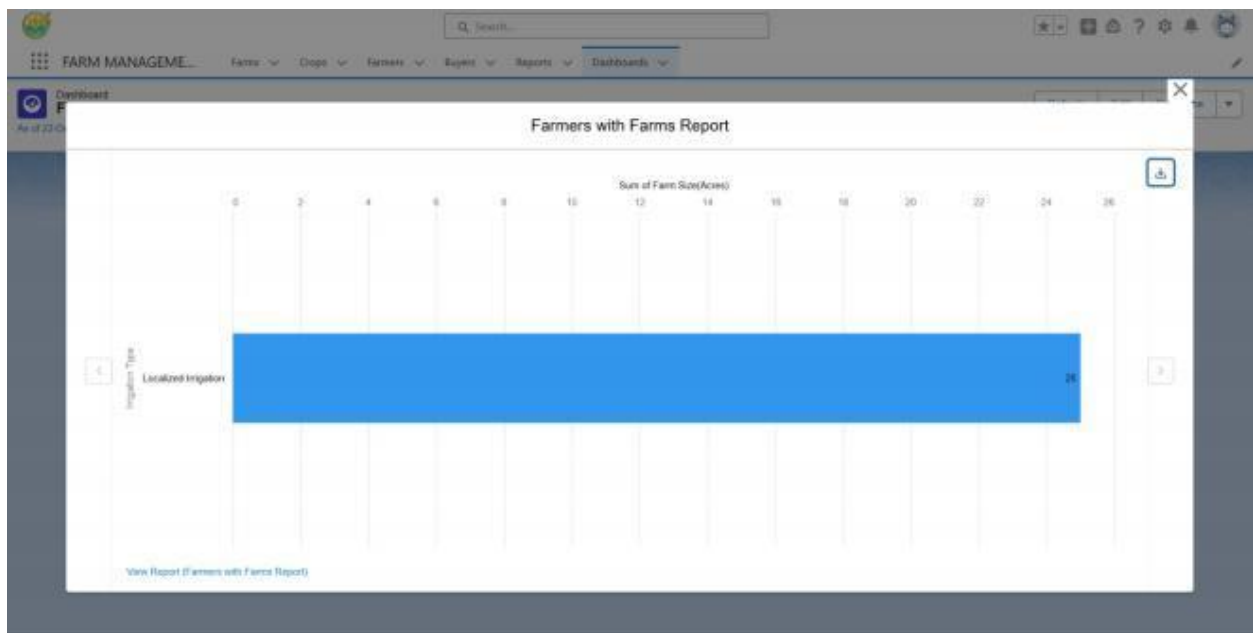


The screenshot displays the 'Farmers with Farms Report' interface. At the top, there is a search bar and navigation tabs for Farms, Crops, Farmers, Buyers, Reports, and Dashboards. The 'Reports' tab is active, showing a sub-report titled 'Farmers with Farms Report'. Below the title, there are summary statistics: 'Total Records: 3' and 'Total Farm Size(Acres): 25'. The main data is presented in a table with columns: Farmer, Farmer Name, Farm, Farm Name, Contact Number, Farm Size(Acres), and Farm Type. The table lists three records: Rajesh with Paddy Farm (10 acres, Livestock), Rajesh with Sugar Farm (10 acres, Crop), and Kishore with Wheat Farm (15 acres, Livestock). A 'Subtotal' row shows a total of 25 acres, and a 'Total (3)' row also shows 25 acres. At the bottom, there are toggle switches for 'Row Counts', 'Detail Rows', 'Subtotals', and 'Grand Total', all of which are currently enabled.

Farmer	Farmer Name	Farm	Farm Name	Contact Number	Farm Size(Acres)	Farm Type
Rajesh		Paddy Farm		999436770	10	Livestock
Rajesh		Sugar Farm		999436770	10	Crop
Kishore		Wheat Farm		999436770	15	Livestock
Subtotal					25	
Total (3)					25	

Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.



7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Relationship and dependency checks
- Profile and role access testing
- Sharing rule verification
- Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data system
- Reduced paperwork
- Improved accuracy
- Better monitoring
- Real-time analytics

Disadvantages

- Requires internet connection
- Salesforce license dependency
- Limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

- Mobile application integration
- AI-based crop prediction
- Weather data integration
- IoT sensor connectivity
- Government scheme integration
- Market price forecasting

11. APPENDIX

- Source Code: Salesforce Configuration and Metadata
- Dataset: Farm, Crop, Farmer, and Buyer Records
- Tools Used: Salesforce Developer Org

Thank You

